

NIBREXO OS AI

12 — FINAL LLM ARENA MASTER BUILD

SPECIFICATION v1.0

Final coding-agent handoff document for building Nibrexo OS AI as a production digital product for Nibrexo.

0. MASTER DIRECTIVE

Build Nibrexo OS AI as a secure centralized operating system for Nibrexo. The product must combine social-media operations, client communication, email automation, content management, CRM and an AI agent into one controlled workspace. Use the approved core stack only: GitHub for version control, Supabase for authentication/database/backend services, Vercel for hosting/deployment/domain, and official APIs for supported external platforms. Do not introduce unnecessary third-party coding or deployment services.

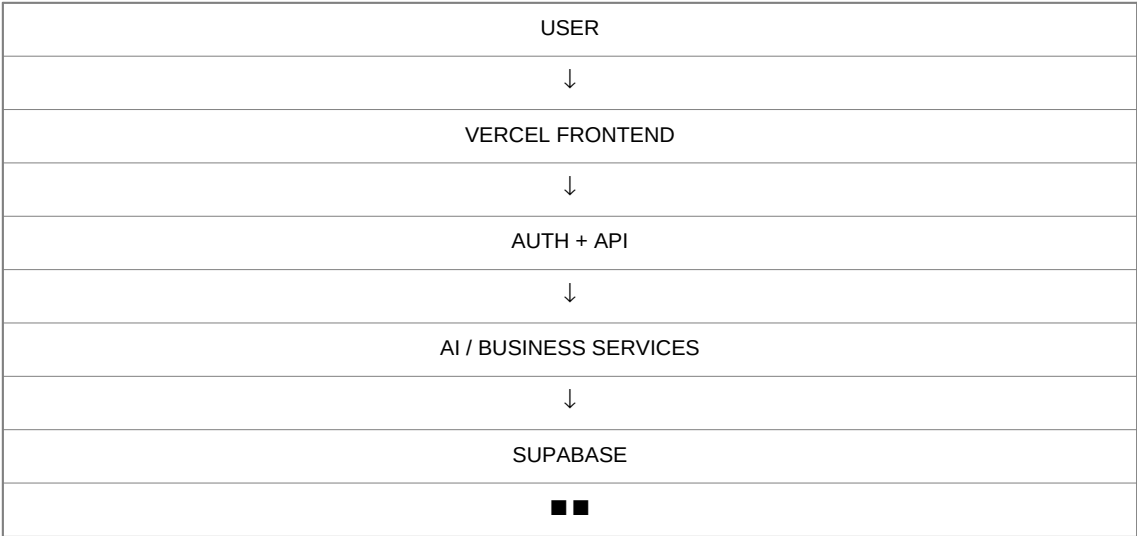
1. Product Vision

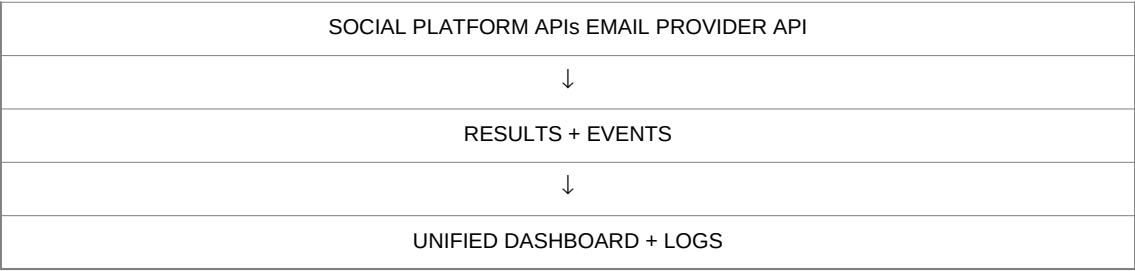
Nibrexo OS AI ka purpose ye hai ke Nibrexo team ko alag-alag platforms aur tools ke darmiyan repeatedly switch na karna pade. Connected social accounts, content publishing, supported DMs/comments, clients, templates, email automation aur AI-assisted operations ko ek centralized dashboard se control kiya ja sake.

2. Core Product Modules

Module	Core Responsibility
Dashboard	Central operational overview
Social	Connected accounts, publishing, scheduling, supported inbox
Content	Media, drafts, captions, publishing assets
Unified Inbox	Supported DMs/comments in one place
Clients / CRM	Client profiles, activities and communication history
Email	Templates, composer, automation and tracking
AI Agent	Planning, tools, memory, approvals and execution
Settings	Organization, accounts, permissions and configuration

3. Master System Flow



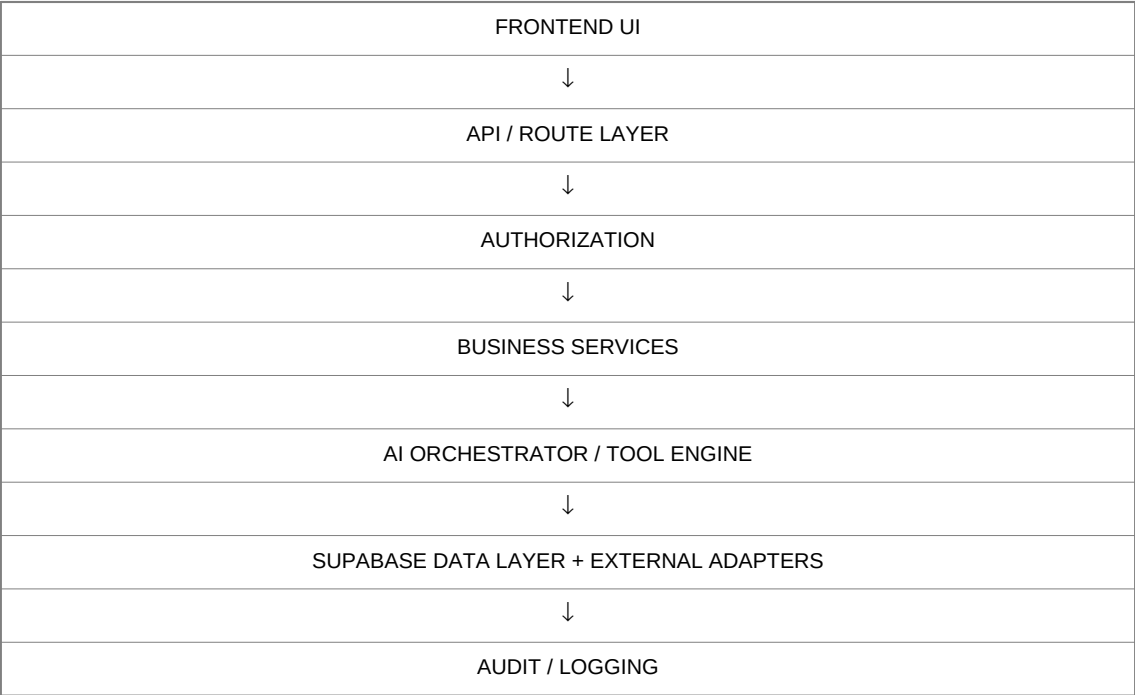


4. Technology Rules

Technology	Required Role
GitHub	Version control, branches, pull requests
Supabase	Auth, PostgreSQL database, RLS and approved backend capabilities
Vercel	Frontend/backend deployment and domain
Official Platform APIs	Social integrations
External Email Provider API	Production email delivery

- No random third-party service should be added simply to avoid implementing a core feature.
- Use official APIs for external platforms.
- Verify current API capability before implementing any platform action.
- Keep provider-specific code isolated behind adapters.

5. Architecture Layers



6. Repository Structure — Recommended

nibrexo-os-ai/ ■■■ app/ or src/app/ ■■■ components/ ■■■ features/ ■■■ dashboard/ ■■■ social/ ■■■ content/ ■■■ inbox/ ■■■ clients/ ■■■ email/ ■■■ ai/ ■■■ settings/ ■■■ server/ ■■■ auth/ ■■■ services/ ■■■ integrations/ ■■■ agent/ ■■■ jobs/ ■■■ lib/ ■■■ supabase/ ■■■ migrations/ ■■■ policies/ ■■■ tests/ ■■■ docs/

7. Database Master Scope

Area	Core Data
Identity	users, organizations, memberships, roles
Social	social_accounts, capabilities, sync state
Content	content_items, media, publishing jobs/results
Inbox	conversations, messages, comments
CRM	clients, contacts, activities
Email	templates, sequences, steps, jobs, logs, events, preferences
AI	agent tasks, tool executions, approvals, memory/context
Audit	activity/audit logs

All organization-owned records must be tenant-scoped and protected by appropriate RLS/server authorization.

8. AI Agent Master Architecture

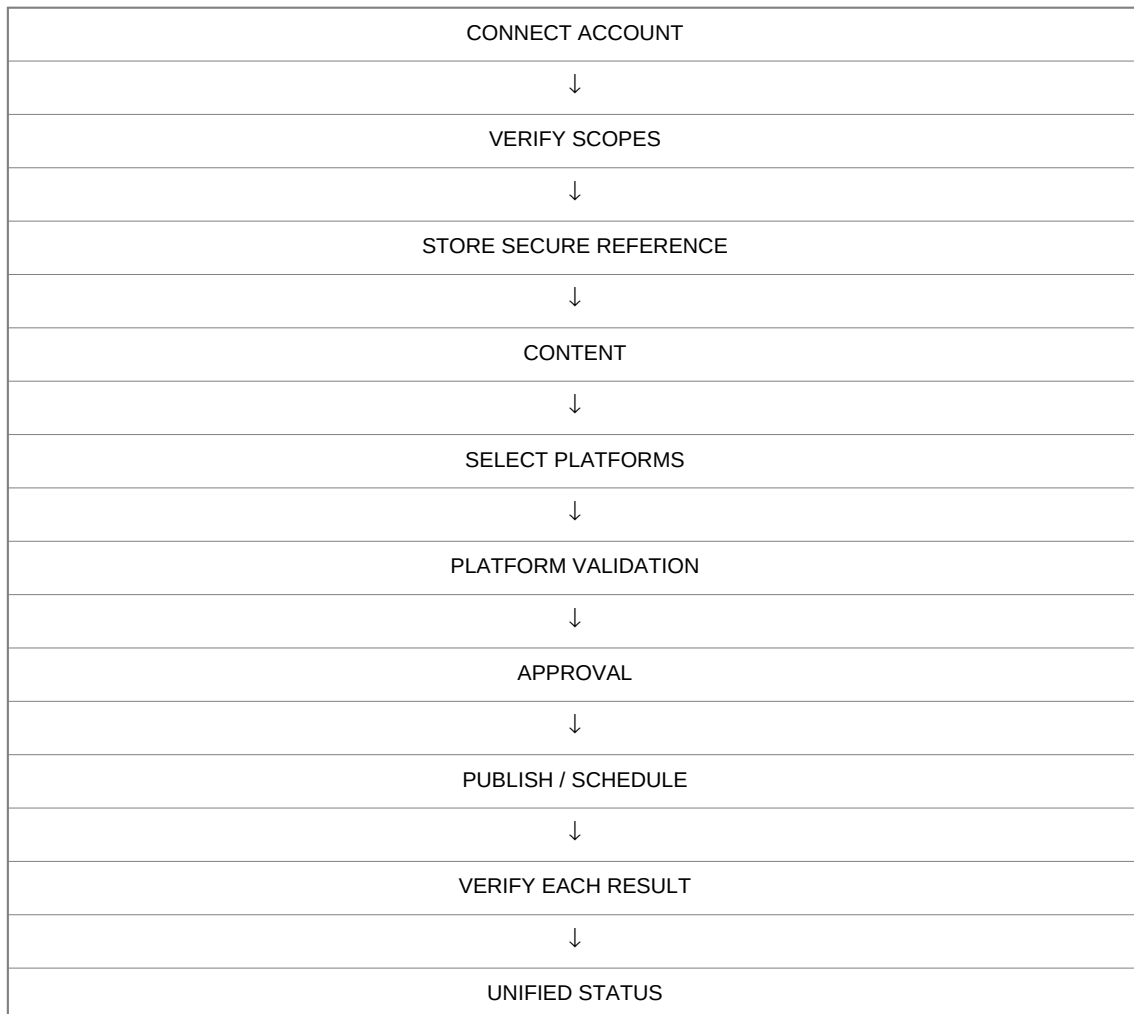


9. Agent Tool Rules

- Tools must have explicit schemas.
- Tools must declare permissions.
- Agent can call only registered tools.

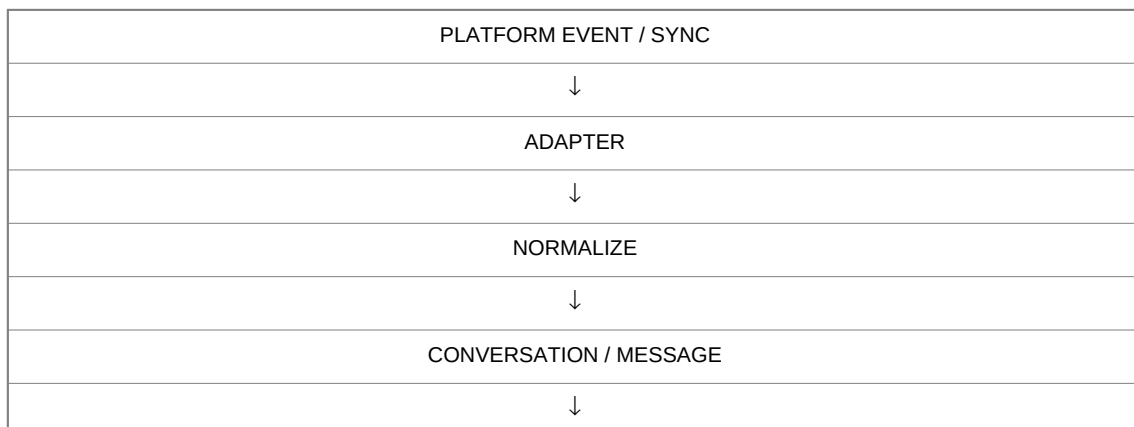
- External actions run server-side.
- Unknown tools are rejected.
- Important actions are logged.
- AI cannot grant itself access.

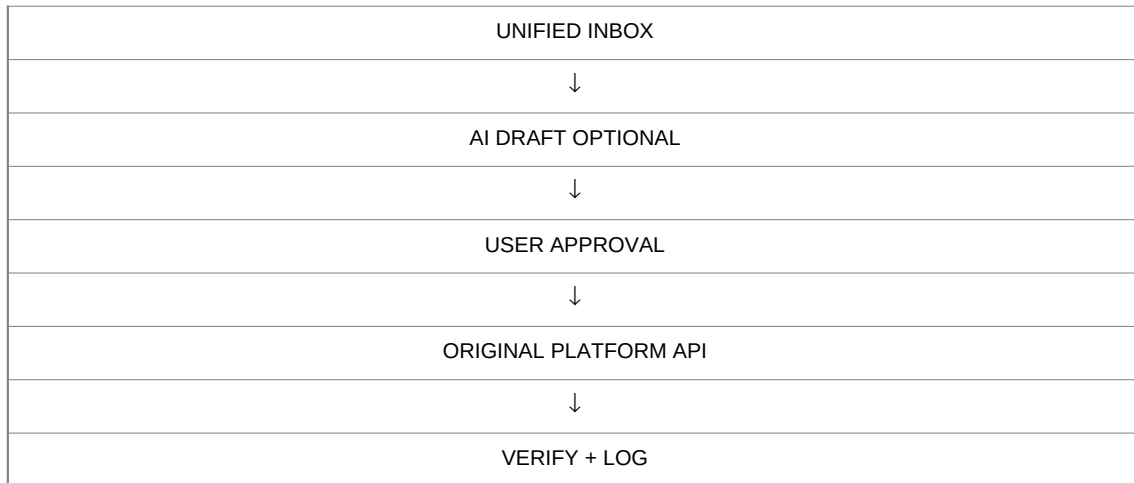
10. Social Master Workflow



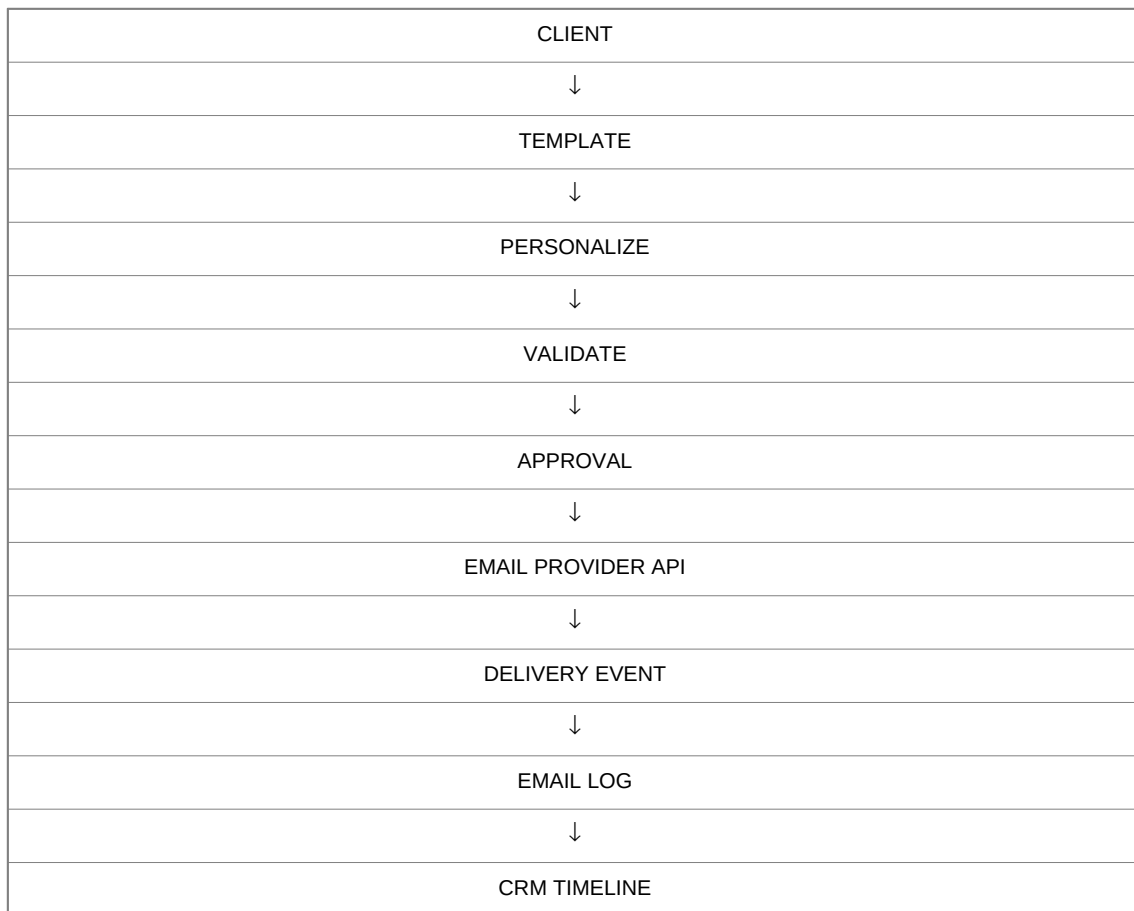
Supported platform targets include TikTok, YouTube, Pinterest, LinkedIn, Instagram, Facebook and Contra only to the extent their current official APIs provide the required capabilities.

11. Unified Inbox Master Workflow

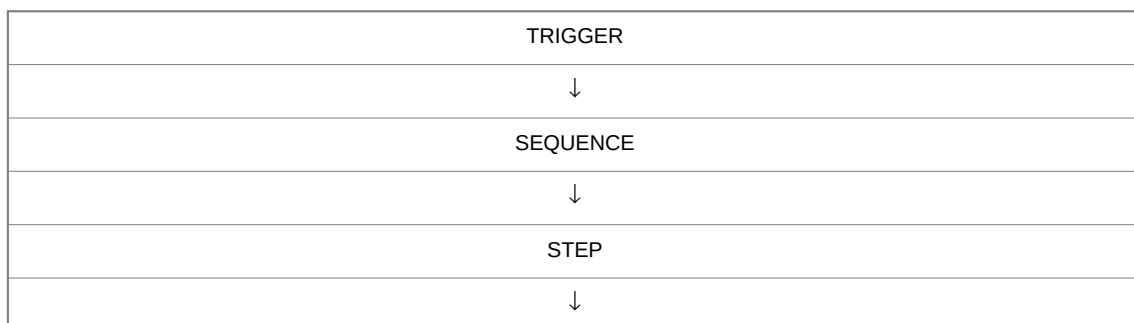




12. Email Master Workflow



13. Email Automation Master Workflow





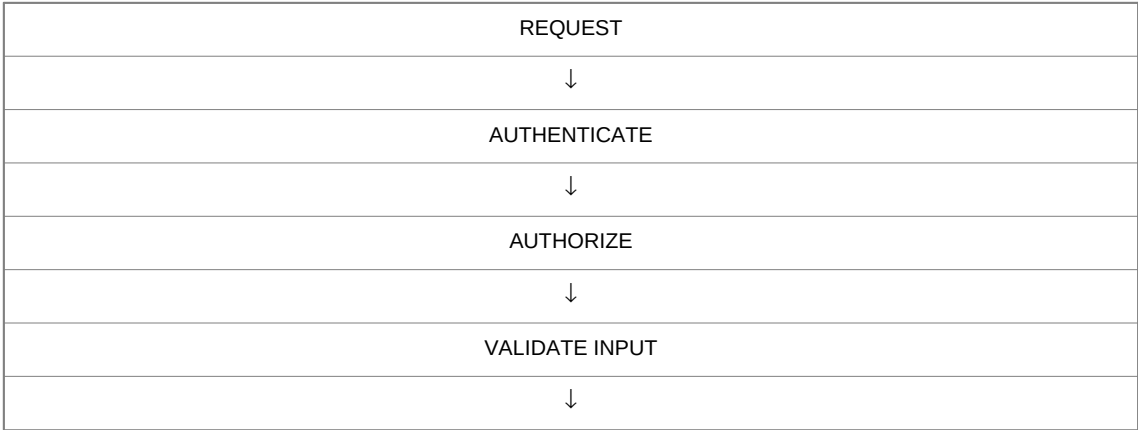
14. Frontend Master Navigation

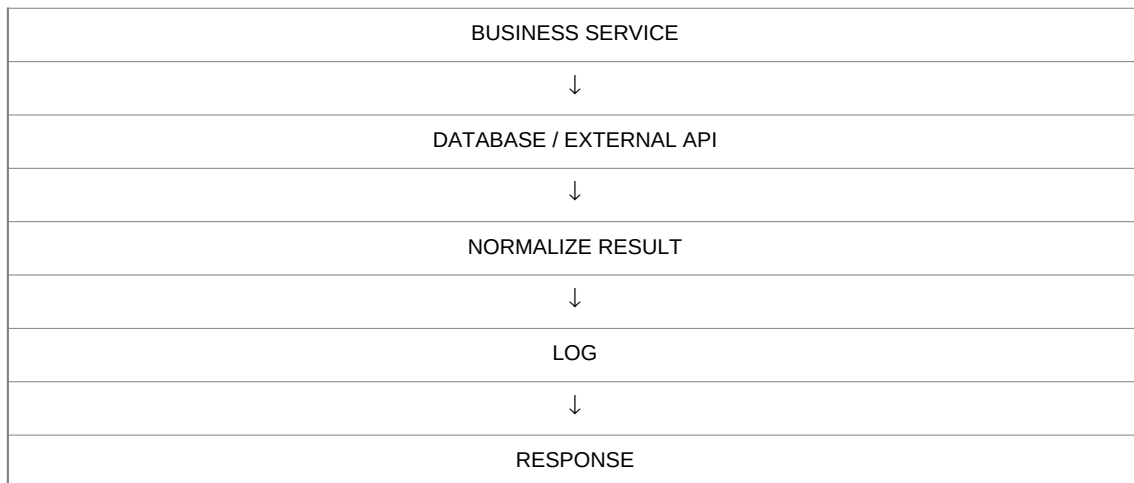
Navigation	Main Screen
Dashboard	Overview and quick actions
Social	Accounts, publishing, calendar
Content	Library and drafts
Inbox	Messages/comments
Clients	CRM and client activity
Email	Templates, composer, sequences, logs
AI	Agent workspace
Settings	Organization, integrations, permissions

15. User Roles & Permissions

- Organization owner/admin controls configuration and permissions.
- Team members receive only required module access.
- Client/resource access must be organization-scoped.
- AI uses the same authorization boundaries as users.
- Frontend hides unavailable actions, but backend remains authoritative.

16. API Design Principles





- Use clear resource-oriented API routes.
- Return structured success/error responses.
- Do not expose internal secrets.
- Validate every external action.
- Make retryable operations idempotent.

17. Security Master Rules

- Supabase RLS protects tenant data.
- Privileged API keys remain server-side.
- OAuth/API credentials are never placed in frontend code.
- Input validation is mandatory.
- Production secrets live in secure environment configuration.
- AI cannot bypass permissions.
- Important external actions are auditable.
- Do not store unnecessary sensitive information in AI memory.

18. Testing Master Matrix

Area	Minimum Coverage
Auth	Login, logout, expiry, access denial
RLS	Cross-tenant isolation
API	Validation, permission, errors
Social	Connect, publish, schedule, failure, retry
Inbox	Sync, duplicate event, reply
Email	Template, send, event, bounce, retry
AI	Planning, tool permission, approval, failure
UI	Critical workflows and responsive states
Deployment	Build, environment, migrations, smoke test

19. Production Deployment Flow



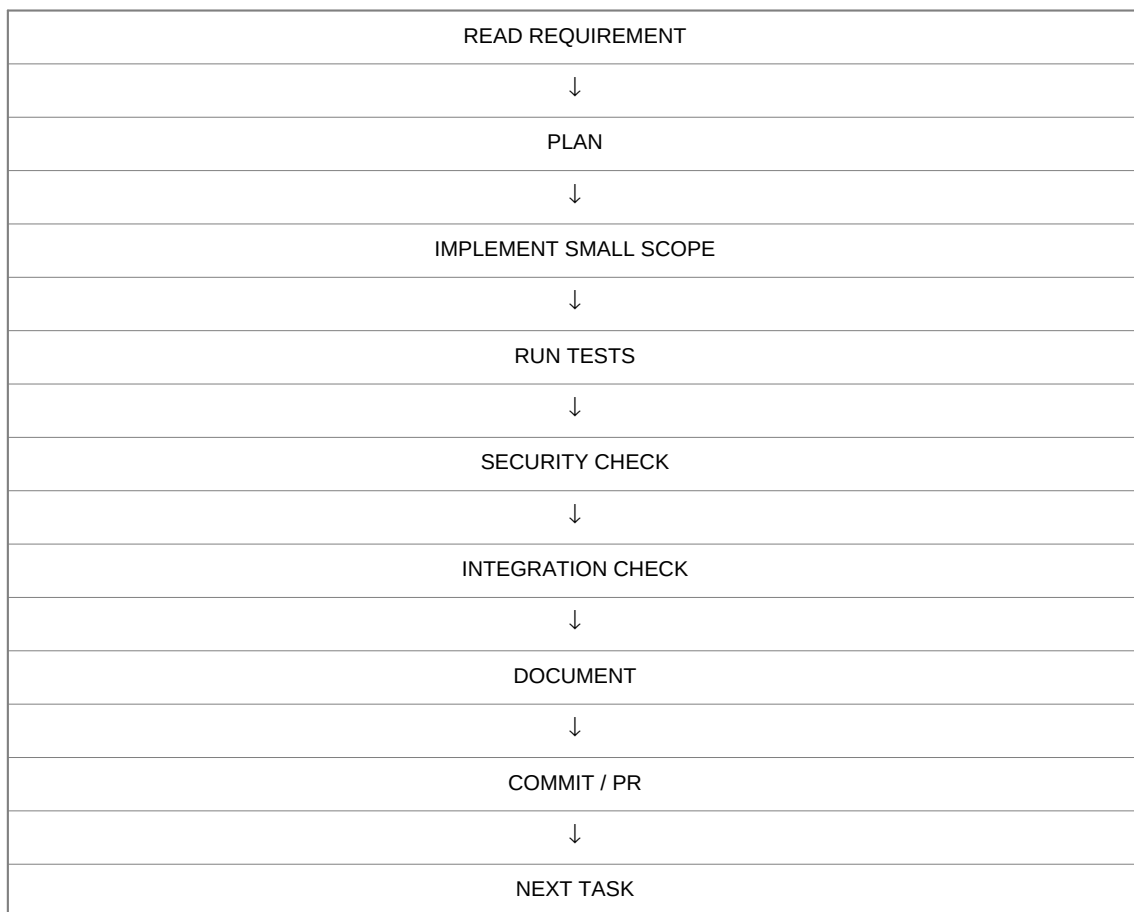
20. Development Roadmap — Build Order

Phase	Build
1	Repository, Supabase project, Vercel project, environment setup
2	Auth, organizations, roles, RLS
3	Database schema + migrations
4	Dashboard shell + navigation + design system
5	Clients/CRM
6	Content library
7	Social account connections
8	Publishing + scheduling
9	Unified inbox
10	Email templates + sending
11	Email sequences
12	AI agent + tool registry
13	Approvals + audit logs
14	Testing, security and production hardening

21. LLM Arena Build Instructions

- First read this entire master specification and the detailed documents.
- Do not start by generating random UI; implement architecture and dependencies in build order.
- Before coding a platform integration, verify the current official API capability.
- Do not invent API endpoints, scopes or unsupported features.
- Create reusable services and adapters rather than duplicating logic.
- After every meaningful module, run tests and verify the feature end-to-end.
- Keep database migrations and security policies version-controlled.
- Do not expose secrets.
- Do not silently add third-party infrastructure.
- Report files changed, migrations added, APIs created and tests run.

22. Agent Coding Loop



23. Definition of Done

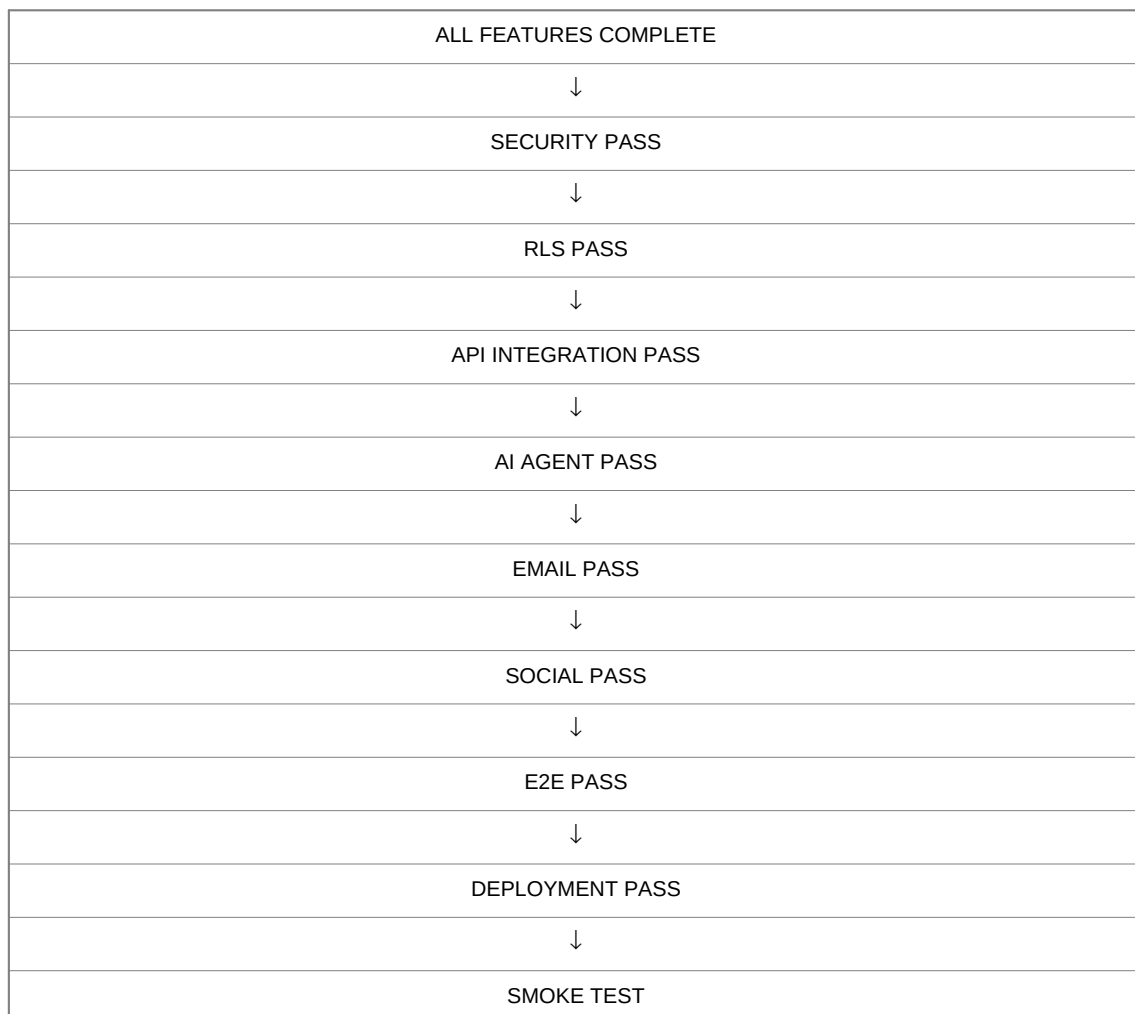
- Feature works through the real frontend-to-backend path.
- Authorization is enforced.
- Database records are tenant-safe.
- Errors have useful user-facing states.
- Important actions are logged.

- External results are verified.
- Retries do not duplicate actions.
- Tests cover success and failure paths.
- Documentation is updated.
- Code is ready for review/deployment.

24. Non-Negotiable Product Rules

- Security over convenience.
- Official APIs over scraping/bypassing.
- Backend authorization over frontend assumptions.
- Verified results over AI claims.
- Human approval where configured.
- Tenant isolation at every layer.
- No secret exposure.
- No uncontrolled autonomous external communication.
- No silent production changes.
- No unnecessary third-party infrastructure.

25. Final Go-Live Flow



↓
GO LIVE ■

26. Documentation Set

#	Document
01–04	Product requirements, technical foundation, roadmap and feature/role specifications
05	Database Architecture & Supabase Schema
06	Backend Architecture & API Design
07	Frontend Architecture & UI/UX System
08	AI Agent Architecture & Automation Engine
09	Social Media Integration & Publishing Specification
10	Email Automation & Client Communication Specification
11	Security, Testing & Production Deployment
12	This Final LLM Arena Master Build Specification

27. Final Instruction to the Coding Agent

Build Nibrexo OS AI as one coherent production system, not as disconnected demos. Preserve the architecture, security boundaries, database tenancy, permission model, adapter pattern, approval system and audit trail described in this specification. When a platform capability is unavailable through its official API, implement a clear unsupported state rather than a workaround that violates platform rules.

■ NIBREXO OS AI — MASTER BUILD
SPECIFICATION COMPLETE

Version 1.0 | Final Developer / LLM Arena Handoff | Nibrexo